

CS & IT ENGINEERING



Introduction to C Programming

Lecture No.1



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TOPICS TO BE
COVERED

.....▶

01 Introduction to 'C'

C Programming

Program :

Google maps : Source

dest

ATM M/c :

Pin

Fully automated
Washing M/c :

Why we need PL?

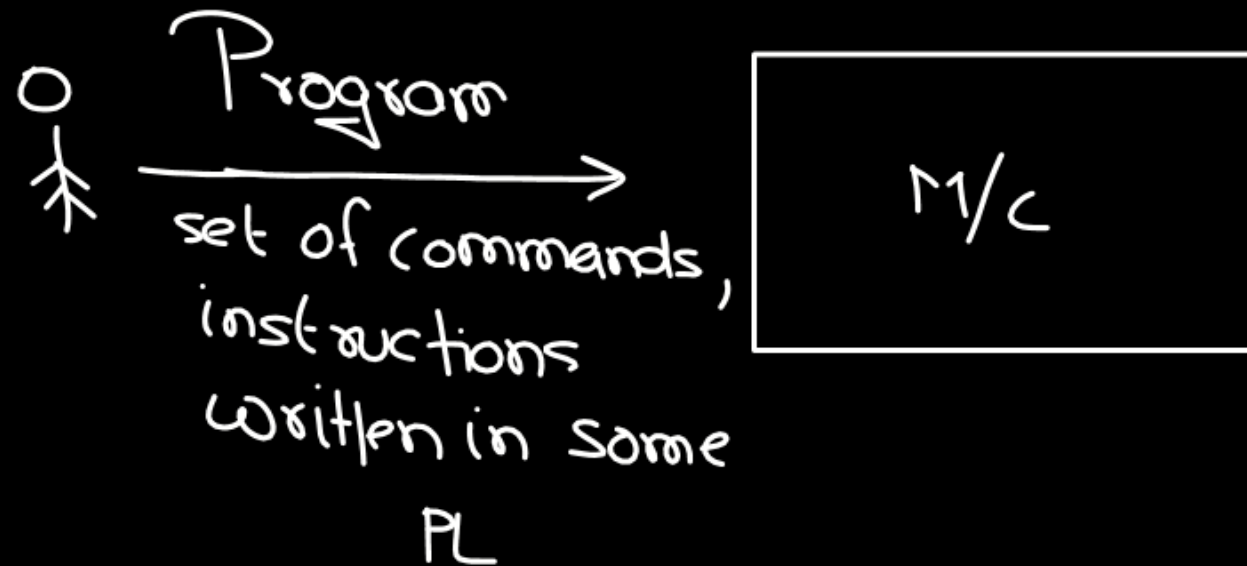


To interact with
m/c we need PL

Why we need to interact with computers?

Computer: dumb

— only follow ^{set of} commands / ^{set of} instruction



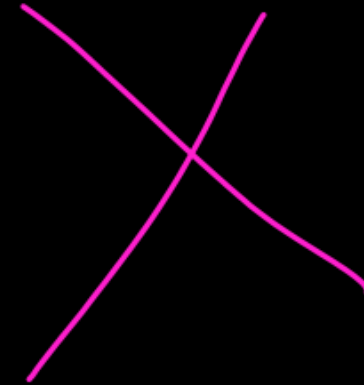

```
#include<stdio.h>
void main(){
    ==
}
```

Can not
directly
give this program
to machine

Computer

English-type language
(Human readable)
Not m/c readable

0,1



```
void main(){  
  
}
```

English type language

Translator

Compiler (best friend)

```
01010--  
10---  
0000..
```

Machine

→ O/p

Can't we write programs
in 0/1 ?

↳ Yes

3, 9, +

00001001 00000011
01001001

M/C language

ADD 3, 9
MUL

vocab

Assembly language
(low level)

a = 3
b = 9
c = a + b

High level
languages



How many lang. ?

Hindi

English

Tamil

Malayalam

Urdu

Marathi

⋮

do i need to
learn all ?

How many PLs ?

C

C++

Python

Java

⋮

Grammar

Ram to agra going

Tense



Present

Syntax

< Noun > < is / are / am > V' + ing

...



①

To
The

leave application

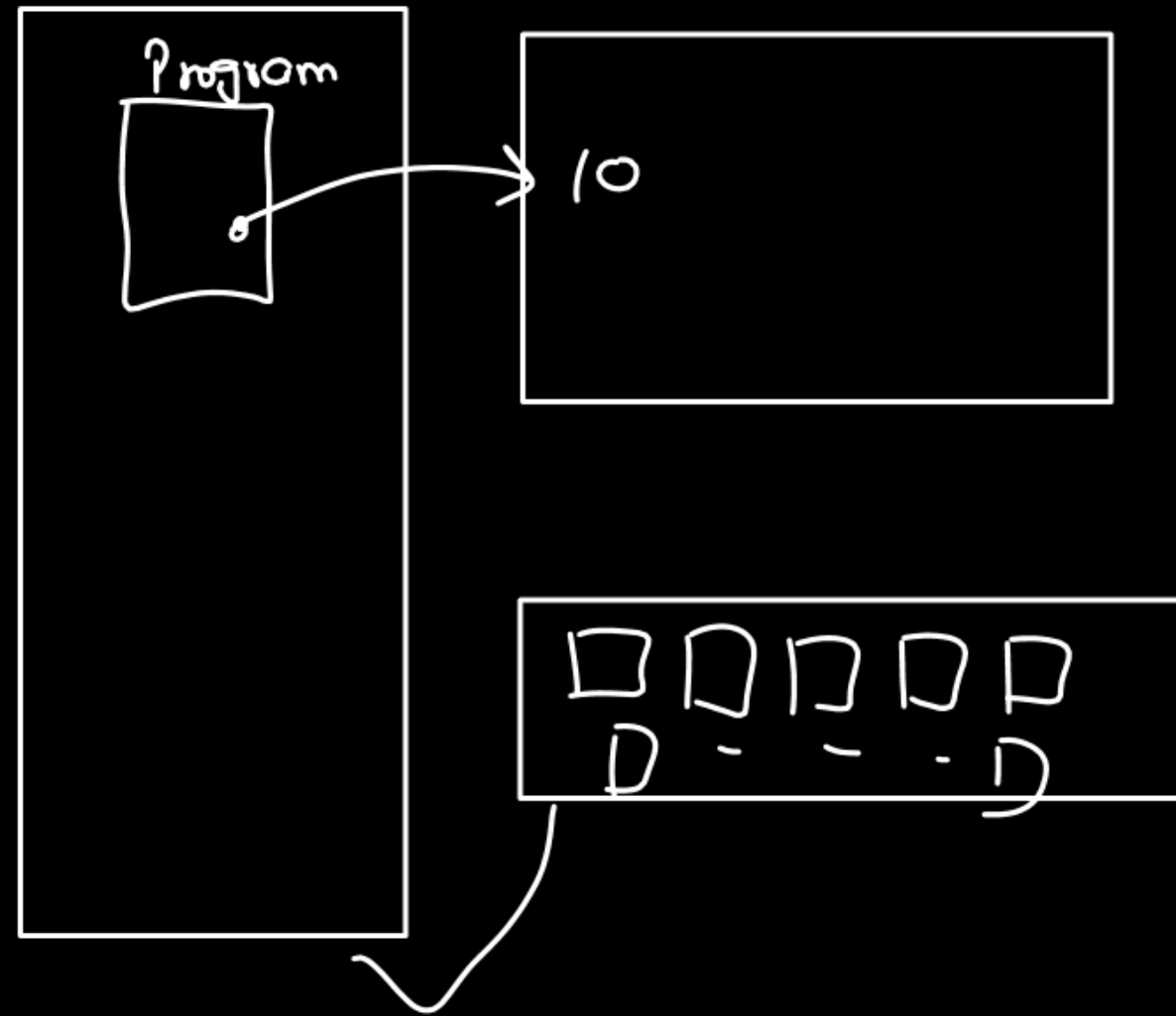
URGENT

Telegram

```
void main(){
```

```
}
```

- ② Every PL must provide the feature to read/take/fetch input coming from the KB.
- ③ Every PL must provide the feature to ^{Print/} give o/p onto monitor.

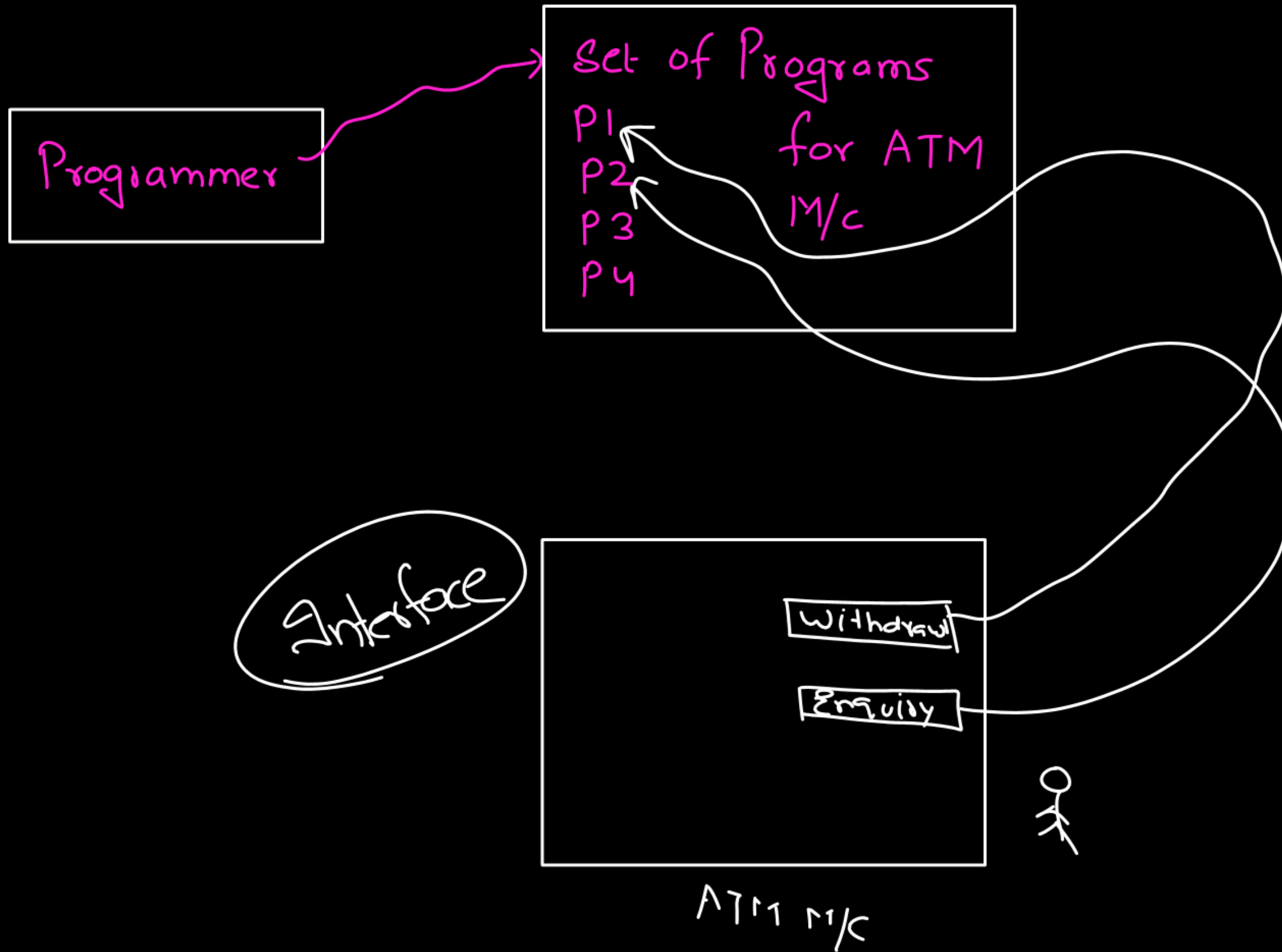


Is it necessary to learn PL
to interact with computers?

Q] Everyone interacting with a comp/m/c ~~is a programmer?~~

1. USER

2. Programmer



S/w

Presentation
(.pptx)
document
(.doc)

Powerpoint

MS word

C Program
(.c)

C
Related
S/w

Dev c++

Turbo C

gcc

Code blocks

VS code

install
S/w

→ Memory

- ① Compiler
- ② Library

Why we need Library?

whatever u provide
in double quotes
it will print it on to
monitor

printf
related
information

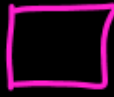
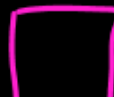
```
#include <stdio.h>
void main() {
```

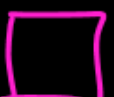
```
    printf("Hello world");
}
```

Hello world

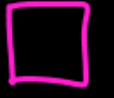
what?

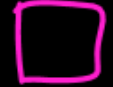
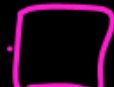
Library

  ... 

  .. 

 . . . 

 . . . 

  . . . 

  . . . 

math.h

stdio.h

variable

being able to vary (change)

$$y = 3x$$

$$x = 1, y = 3$$

$$x = 2, y = 6$$

$$x = 3, y = 9$$



Variable — data/value

Store value
10

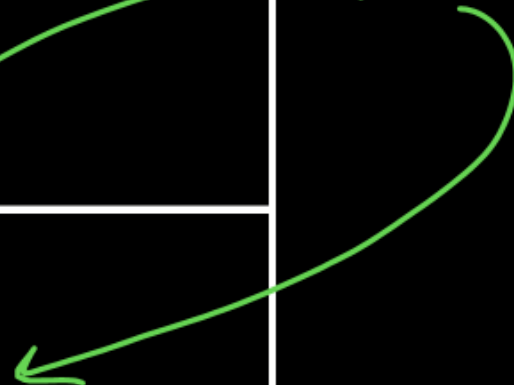
How to retrieve?

  
Father Mother

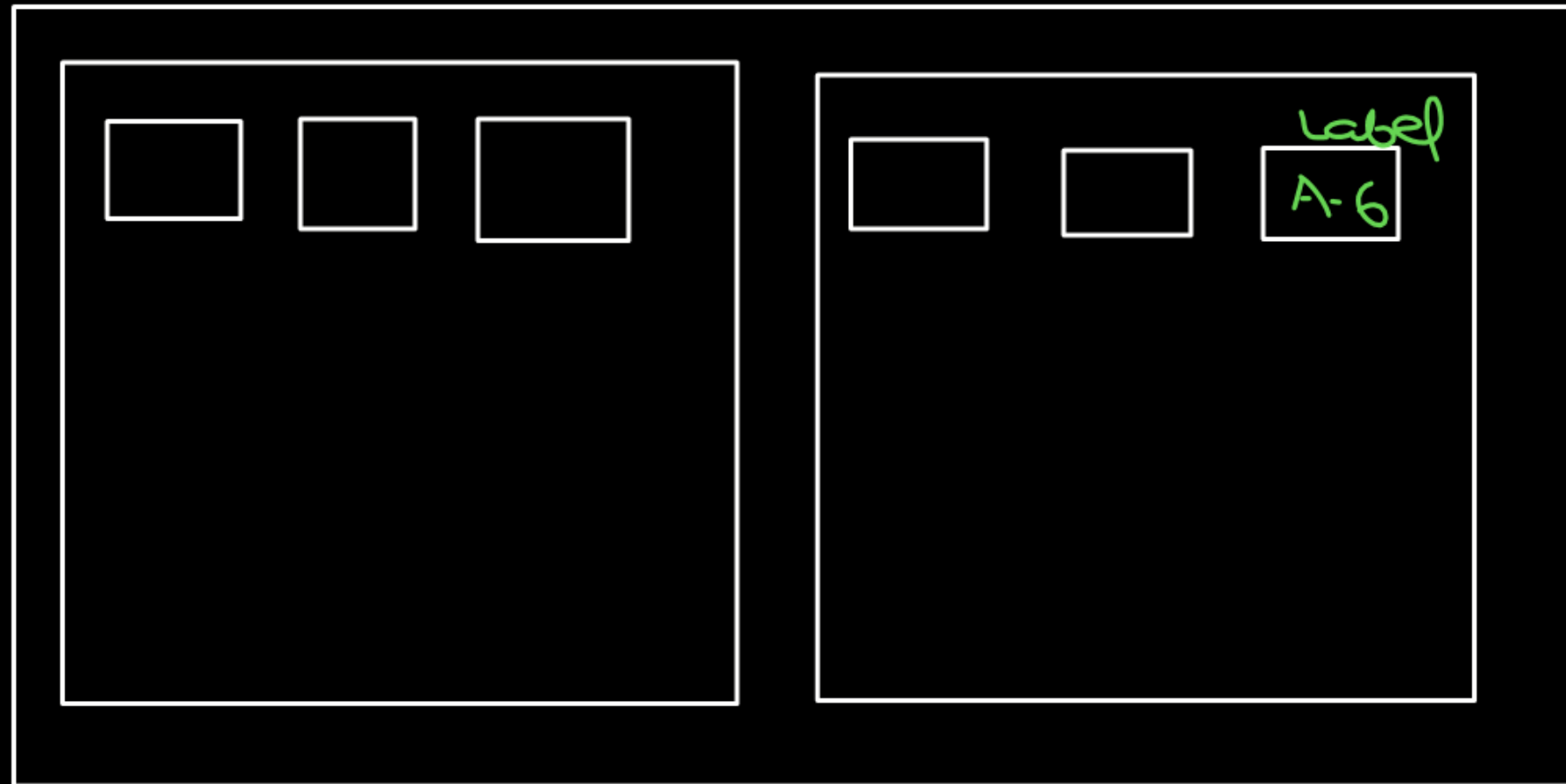
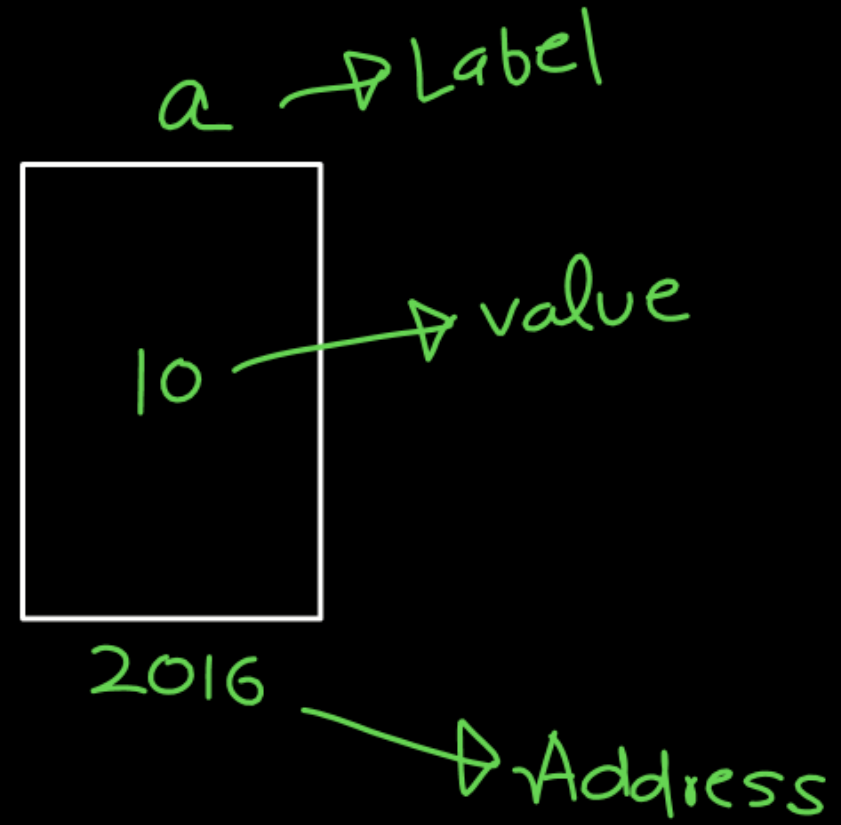
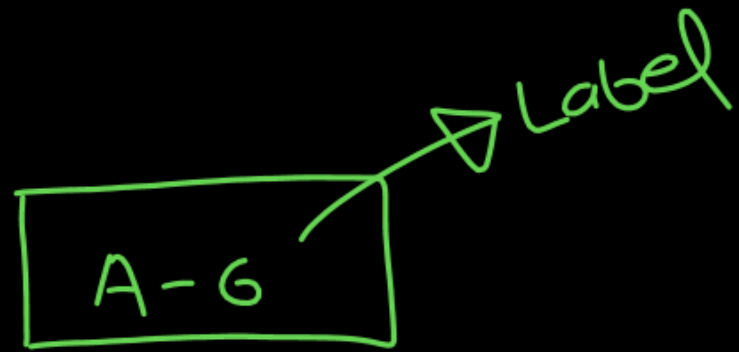


1198	a	
		

Identity



$a = 10$



Google maps

Source

dest.

X

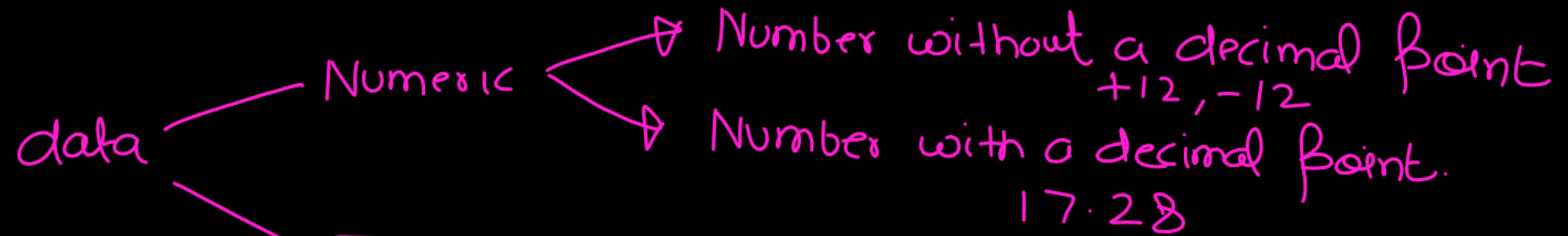
Goa ✓
Delhi ✓ } textual



ATM M/c

Pin

Numeric



Text : address :
mail :

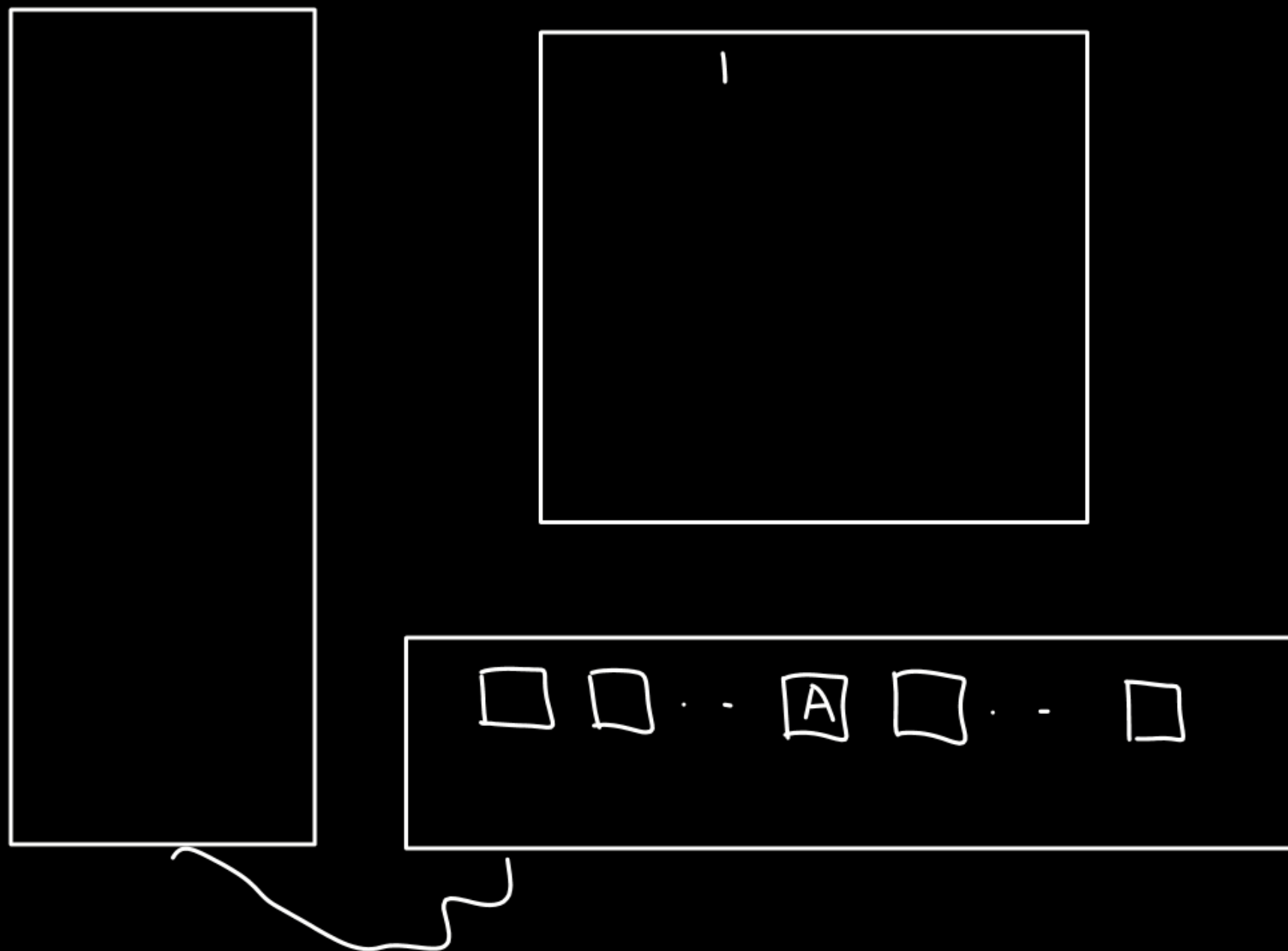
Numbers

$$10 + 20 = ?$$

$$10 + \text{"address"} = ?$$

number

Not a no.



whatever we are
providing from key-
board, for m/c
it is a seq. of
0 and 1

$A \rightarrow 01000001$
 ↓
 65

